# Exploratory Data Analysis On India's Amenities, Buildings and Leisure Places.

# Overall Summary:

- This presentation explores the datasets involving India’s Amenities, Buildings and Leisure places.
- Various pre-processing had to be done before analysis.
- The Amenity dataset provided insight into the amenities that are most prevalent in India and their location.
- The Buildings dataset provided an insight into the religious sites of the 3 main religions in India.
- The leisure dataset provided and insight into the most prevalent leisure places in India.

# Data Science Questions:

Amenities Dataset:

- What are the top amenities in India?
- How evenly are schools spread or build around the country?
- What does it say about the literacy rate?

Buildings Dataset:

- What are the main religions practiced in India?
- How evenly are their holy sites spread across the country?

Leisure Dataset:

- What are the top leisure laces in the country?
- Are stadiums around the country evenly spread considering the presence of India in world sports.

# Methodology:

The main tool used for this analysis python along with its API PySpark. Other libraries such as numpy, pandas, matplotlib and seaborn were also used.

But before we visualize, we need to do some pre-processing. Most of the pre-processing for the 3 datasets were same.

# Methodology: Pre-Processing

## 1. Adding a column name to the unnamed column.

```
amenity_log.describe().show()
```

```
+-------+------------------+---------------------------
|summary|               _c0|
+-------+------------------+---------------------------
|  count|            552667|
|   mean|5.202562898620065E9|                          5
| stddev|3.077197210317241E9|                172929.179
|    min|         100065924|                'amenity': '
|    max|         arnataka'|V G  K U L K A R I N I  H O S
+-------+------------------+---------------------------
```

ng a name to the unnamed column.

```
amenity_log = amenity_log.withColumnRenamed("_c0", "id")
```

```
[11] leisure_log = leisure_log.withColumnRenamed("_c0", "id")
```

```
leisure_log.show()
```

```
+---------+-------------------+------------+-------------------+-------------------
|       id|               name|     leisure| longitude-lattitude|           All_tags|
+---------+-------------------+------------+-------------------+-------------------
```

```
[21] building_log = building_log.withColumnRenamed("_c0", "id")
```

# Methodology: Pre-Processing

**2. Finding the null values. All the null values are in the longitude-lattitude and name columns.**

```
from pyspark.sql.functions import isnan,when,count

null_count = amenity_log.select([count(when(col(c).contains('None') | \
                                 col(c).contains('NULL') | \
                                 (col(c) == '' ) | \
                                 col(c).isNull() | \
                                 isnan(c), c
                                )).alias(c)
                       for c in amenity_log.columns])
```

```
[ ] null_count.show()

+---+-----+-------+------------------+--------+
| id| name|amenity|longitude-lattitude|All_tags|
+---+-----+-------+------------------+--------+
|  0|50136|      0|             68337|       6|
+---+-----+-------+------------------+--------+
```

```
null_count = building_log.select([count(when(col(c).contains('None') | \
                                 col(c  (variable) c: Any | \
                                 (col(c) == '' ) | \
                                 col(c).isNull() | \
                                 isnan(c), c
                                )).alias(c)
                       for c in building_log.columns])
```

```
null_count.show()

+---+-------+--------+------------------+--------+
| id|   name|building|longitude-lattitude|All_tags|
+---+-------+--------+------------------+--------+
|  0|9896644|       0|           9979088|       2|
+---+-------+--------+------------------+--------+
```

```
null_count = leisure_log.select([count(when(col(c).contains('None') | \
                                 col(c).contains('NULL') | \
                                 (col(c) == '' ) | \
                                 col(c).isNull() | \
                                 isnan(c), c
                                )).alias(c)
                       for c in leisure_log.columns])
```

```
[14] null_count.show()

+---+-----+-------+------------------+--------+
| id| name|leisure|longitude-lattitude|All_tags|
+---+-----+-------+------------------+--------+
|  0|27143|      0|             37876|       2|
+---+-----+-------+------------------+--------+
```

# Methodology: Pre-Processing

**3. Removing the null values of longitude-latitude column only as the name column may not have been recorded due to dozens of languages being present in India.**

No. of entries in longitude-lattitude column before dropping.

```
[ ] amenity_log.select("longitude-lattitude").count()
```

311462

Dropping duplicates

```
[ ] amenity_log_2 = amenity_log.na.drop(subset=["longitude-lattitude"]
```

```
amenity_log_2.show()
```

No. of entries in longitude-lattitude column after dropping.

```
[ ] amenity_log_2.select("longitude-lattitude").count()
```

243125

No. of entries in longitude-lattitude column before dropping.

```
[16] leisure_log.select("longitude-lattitude").count()
```

43689

Dropping the null values.

```
leisure_log_2 = leisure_log.na.drop(subset=["longitude-lattitude"])
```

```
[18] leisure_log_2.select("longitude-lattitude").count()
```

5813

No. of entries in longitude-lattitude column before dropping.

```
building_log.select("longitude-lattitude").count()
```

10012423

Dropping null values.

```
[25] building_log_2 = building_log.na.drop(subset=["longitude-lattitude"]
```

```
building_log_2.select("longitude-lattitude").count()
```

33335

# Methodology: Pre-Processing

## 4. Dropping duplicates from the longitude-latitude columns.

```
[ ] amenity_log_2.select("longitude-lattitude").count()

243125

[ ] amenity_log_2.select("longitude-lattitude").distinct().count()

243091
```

Dropping locations with same co-ordinates.

```
[ ] amenity_log_3 = amenity_log_2.dropDuplicates(["longitude-lattitude"])
```

```
building_log_2.select("longitude-lattitude").distinct().count()

33320
```

Dropping locations with same co-ordinates.

```
[30] building_log_3 = building_log_2.dropDuplicates(["longitude-lattitude"])

[31] building_log_3.select("longitude-lattitude").distinct().count()

33320
```

```
[19] leisure_log_2.select("longitude-lattitude").distinct().count()

5811

leisure_log_3 = leisure_log_2.dropDuplicates(["longitude-lattitude"]

[21] leisure_log_3.select("longitude-lattitude").count()

5811
```

# Methodology: Pre-Processing

## 5. Removing brackets from the longitude-latitude columns.

Remove brackets from the longitude-lattitude column.

```
[ ] amenity_log_4 = amenity_log_3.withColumn("longitude-lattitude", F.regexp_replace(F.regexp_replace(F.regexp_replace("longitude-lattitude", "\\)\\(", ""), "\\(", ""), "\\)", ""))
```

```
[32] building_log_4 = building_log_3.withColumn("longitude-lattitude", F.regexp_replace(F.regexp_replace(F.regexp_replace("longitude-lattitude", "\\)\\(", ""), "\\(", ""), "\\)", ""))
```

```
[22] leisure_log_4 = leisure_log_3.withColumn("longitude-lattitude", F.regexp_replace(F.regexp_replace(F.regexp_replace("longitude-lattitude", "\\)\\(", ""), "\\(", ""), "\\)", ""))
```

# Methodology: Pre-Processing

## 6. Splitting the longitude-lattitude column into longitude and lattitude and dropping the original.

```
from pyspark.sql.functions import split

new_building_log = building_log_4.withColumn('longitude', split(building_log_4['longitude-lattitude'], ',') .getItem(0)) \
                        .withColumn('lattitude', split(building_log_4['longitude-lattitude'], ',') .getItem(1))

new_building_log.show()

+----------+--------------------+--------+--------------------+--------------------+------------------+--------------------+
|        id|                name|building| longitude-lattitude|            All_tags|         longitude|           lattitude|
```

```
new_building_log = new_building_log.drop("longitude-lattitude")

[38] new_building_log.show()

+----------+--------------------+--------+--------------------+------------------+--------------------+
|        id|                name|building|            All_tags|         longitude|           lattitude|
```

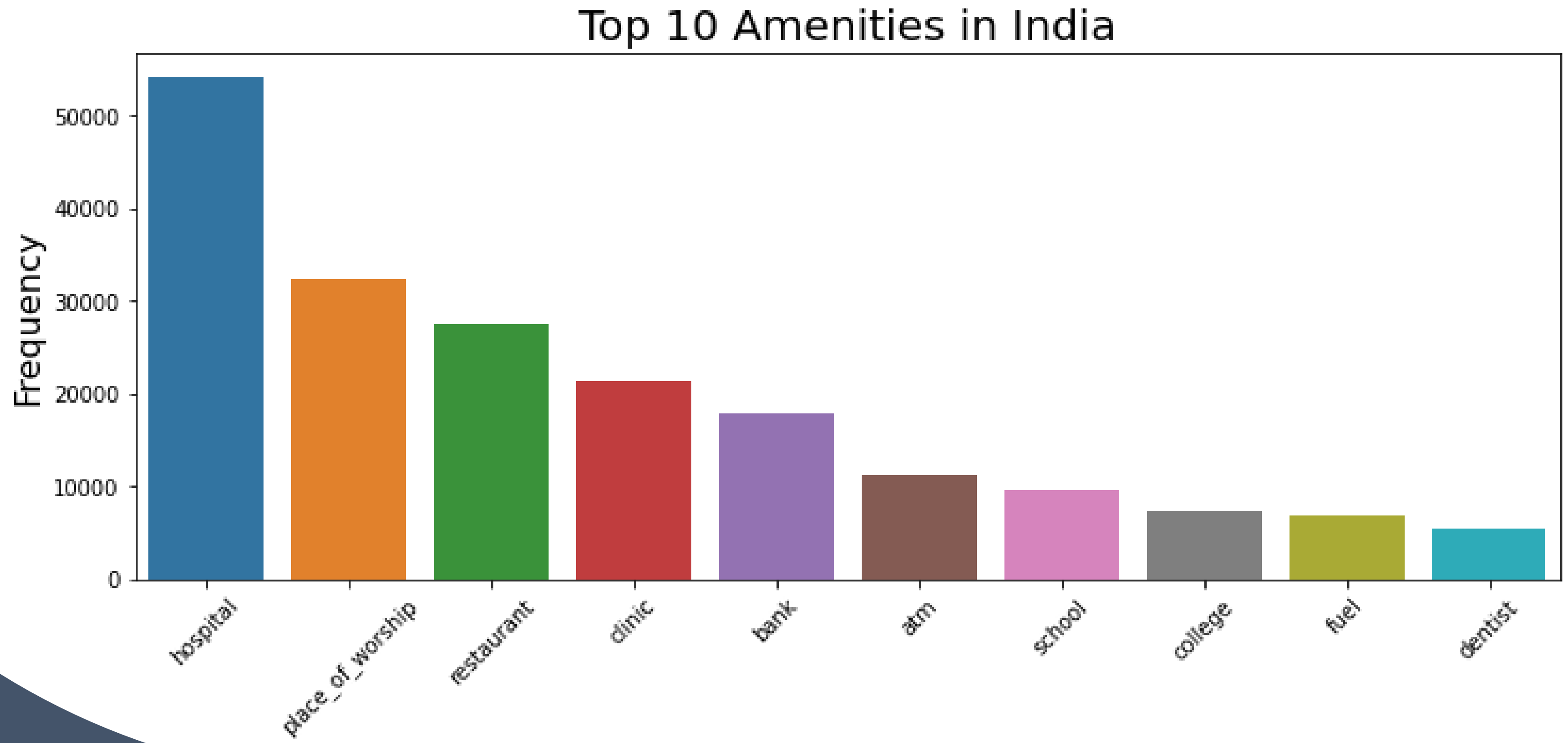
```
[24] from pyspark.sql.functions import split

new_leisure_log = leisure_log_4.withColumn('longitude', split(leisure_log_4['longitude-lattitude'], ',
                        .withColumn('lattitude', split(leisure_log_4['longitude-lattitude'], ',')

[27] new_leisure_log = new_leisure_log.drop("longitude-lattitude")
```

```
from pyspark.sql.functions import split

[ ] new_amenity_log = amenity_log_4.withColumn('longitude', split(amenity_log_4['longitude-lattitude'], ',') .getItem(0)) \
                        .withColumn('lattitude', split(amenity_log_4['longitude-lattitude'], ',') .getItem(1))

new_amenity_log = new_amenity_log.drop("longitude-lattitude")
```

# Results & Visualization: Amenities

We can see that 3 of the top ten amenities recorded are related to health. This is expected since increase in population also means a surplus of medical care that is required.

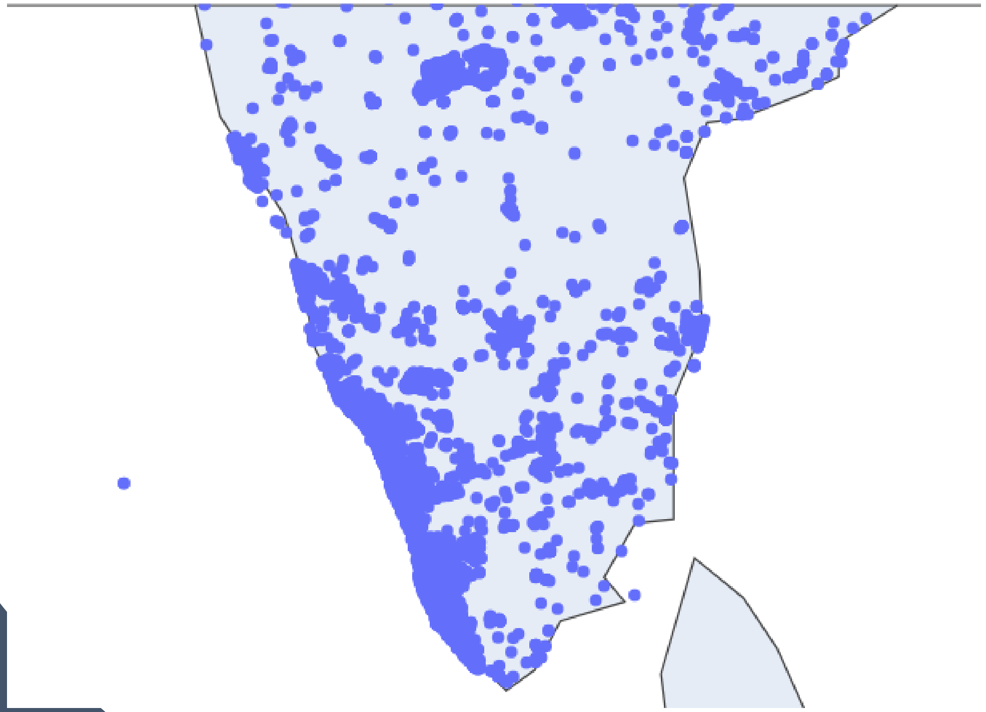
The 2nd most frequent amenity is the place of worship amenity showing that the people of India are religious in mind. We will look into the main religions of India further ahead.



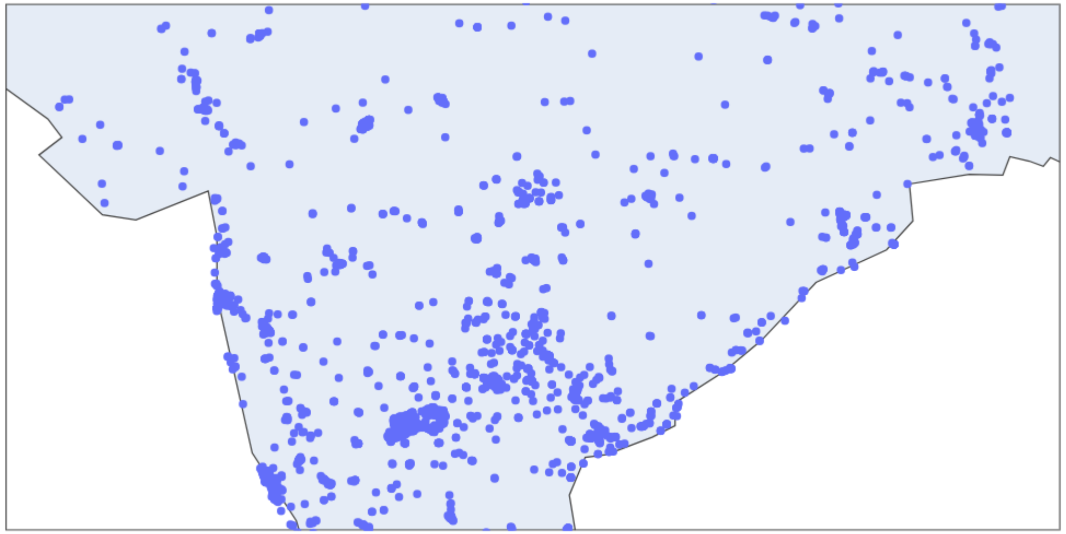
# Results & Visualization: Amenities

If we look into the distribution of schools on a map, we can see that most of the schools are concentrated on the South of the Country, while the central part of the country does not have as much. This does support the fact that the south India boasts the highest literacy rate in the country.

Southern part of India



Central part of India



# Results & Visualization: Buildings

Tallying the frequency of the buildings gives us the highest type of building being “yes” which is misleading. However, if we look at the other entries, we can see that there are 3 religious buildings. Those are the mosques, temples, and churches or chapels. This could very well reveal that the top 3 religions in India are Islam, Hinduism and Christianity, although they may nit say which is the majority.

```
building_frequency = spark.sql('''
                    SELECT building, count(*) as Frequency
                    FROM building_log_table
                    GROUP BY building
                    ORDER by Frequency DESC;
                    '''
                     )
```

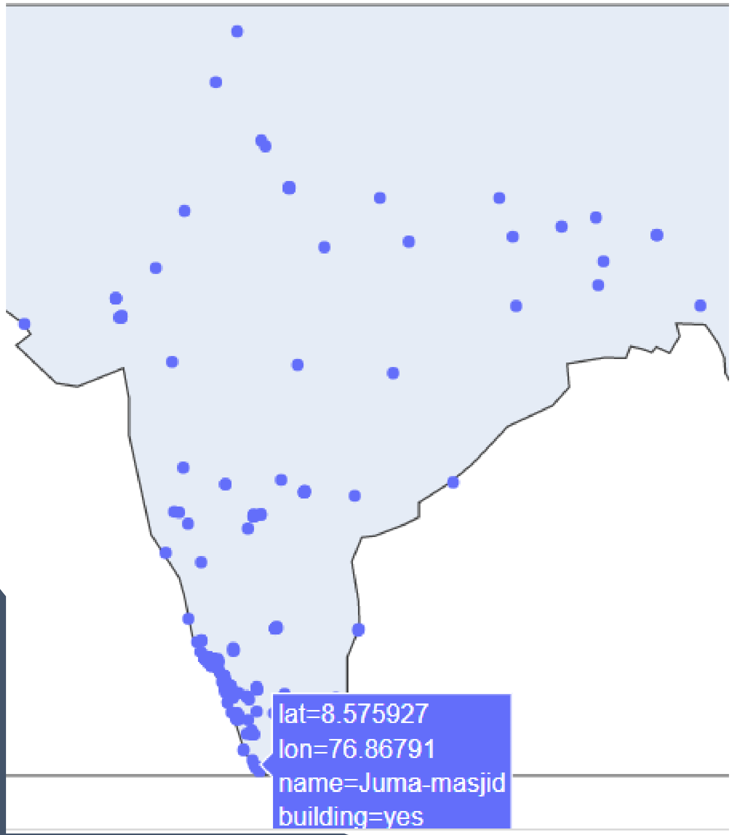
```
building_frequency.show()
```

```
+-----------+---------+
|   building|Frequency|
+-----------+---------+
|        yes|    17741|
|      house|    11692|
|     school|      600|
| apartments|      537|
|residential|      522|
| commercial|      217|
|     office|      215|
|     mosque|      205|
|     temple|      189|
|    college|      152|
|   hospital|      136|
|     public|      135|
| industrial|      133|
|     church|       95|
|       roof|       94|
|     retail|       50|
| university|       48|
|     garage|       44|
|     chapel|       34|
|      hotel|       31|
+-----------+---------+
```

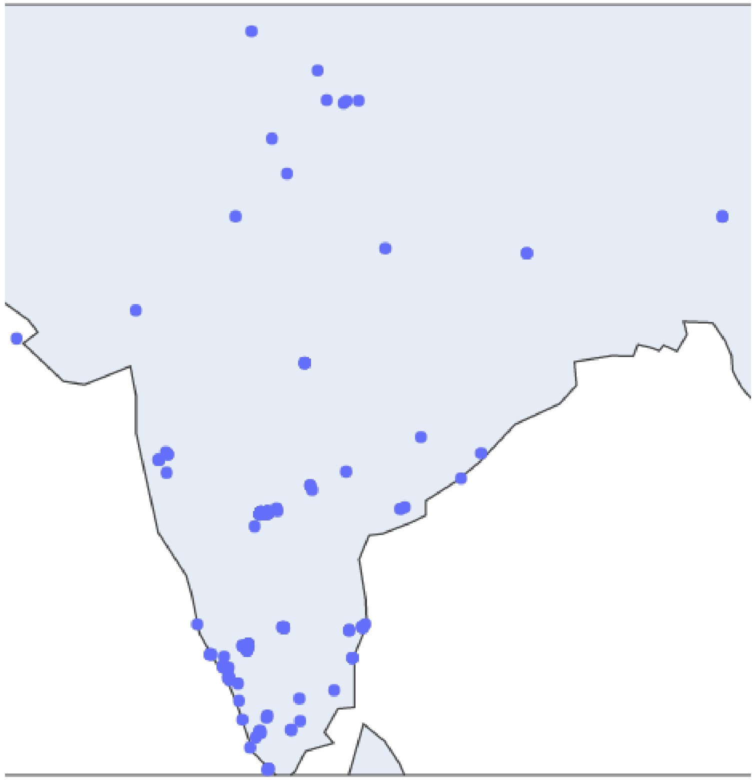
# Results & Visualization: Buildings

Plotting the coordinates of the Mosques, temples and churches, one clear thing we can notice is that there is a concentration of them in the South-West. This could be due to the fact that the data was recorded more clearly in the south. We can also notice a scarcity of churches and temples in central-eastern part of the country.

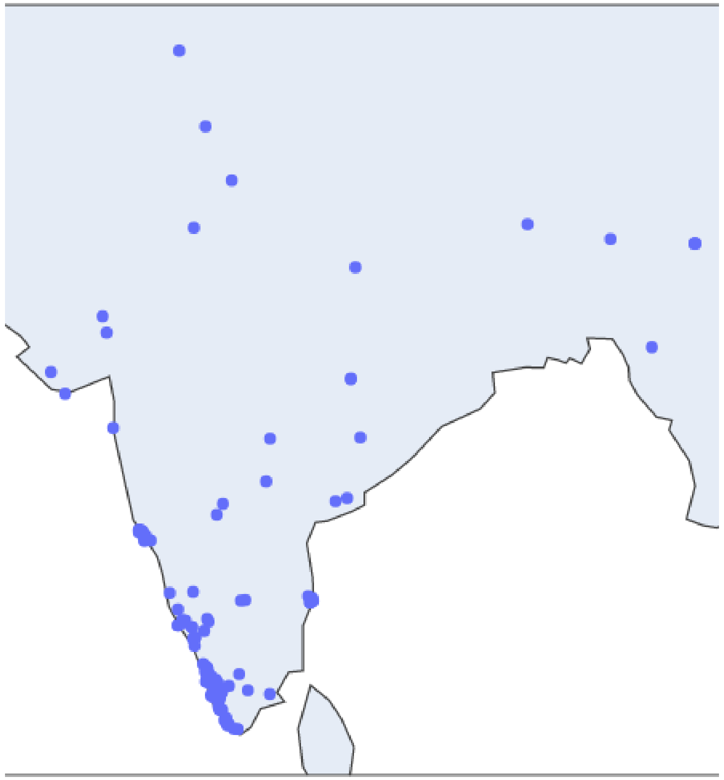
*Mosques*




*Temples*

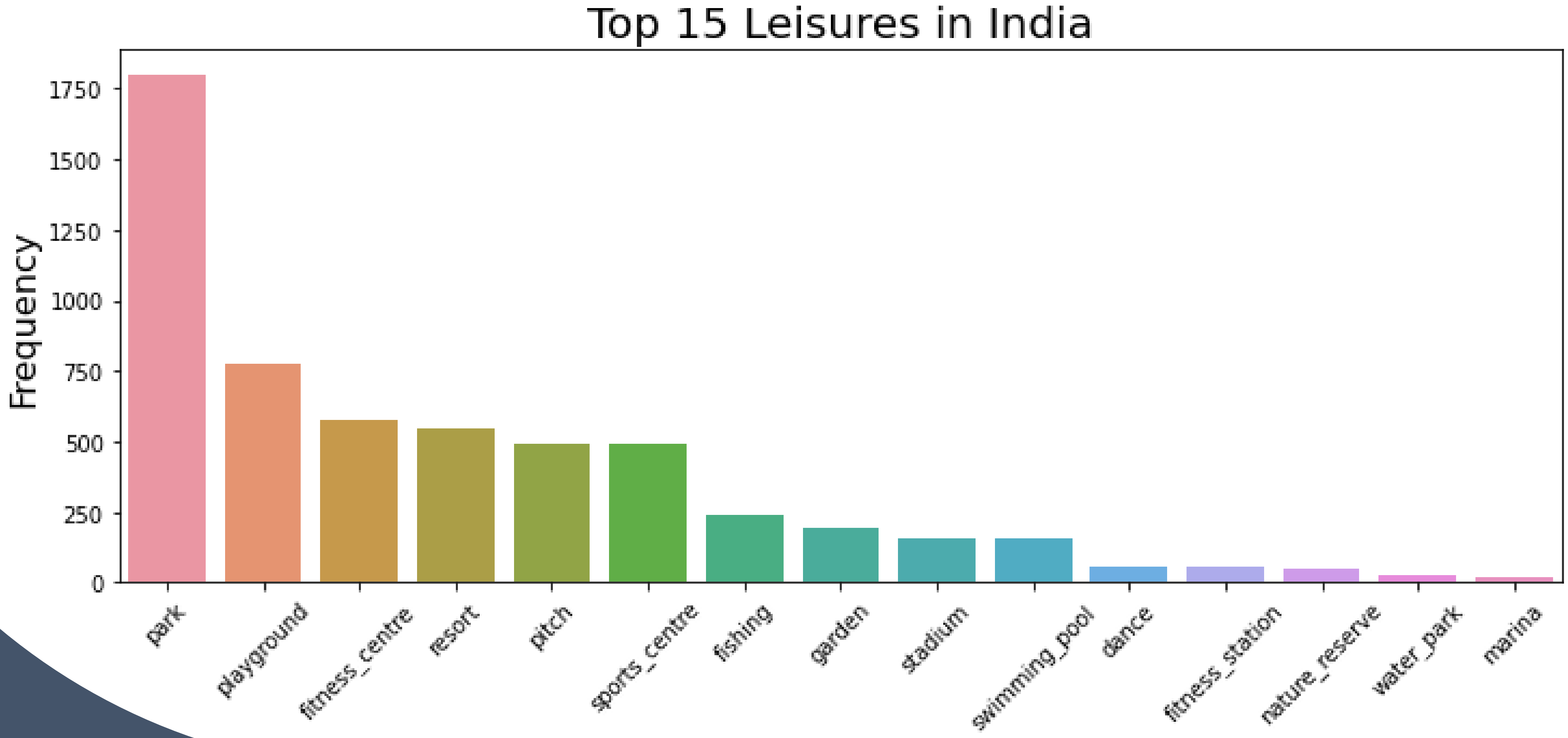


Churches/Chapels



# Results & Visualization: Leisures

The leisure place that has been most recorded are parks that is expectable as India is a sub-tropical country where flora thrives quite easily. Most of the leisure places involve physical activities which tells us that the population is active in nature. The notable leisure that does not involve physical activity seems to be their resorts.



# Results & Visualization: Leisures

India is a country that have a dominant standing in the world of sports, especially Cricket. As such it is expected that it will have many stadiums all across the country. If we look inti the spread of stadiums, we can find that it is quite uneven. Most of the stadiums are in the north or south leaving the central part without much opportunities for spectacles in said stadiums.

*Stadiums*

# Thank you